

Assignment 1

Causal Inference

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1 Descriptive Analysis

1.1 Summarizing data

Data presents a **total number of units** is $N = 60$ defined over $p = 6$ variables representing a **randomized experiment** with $N_t = 32$ **treatment** units and $N_c = 28$ **control** units.

Table 1 shows descriptive statistics for numerical variables. *Sex* variable is categorical and the number of units by sex is shown in 2.

Descriptive statistics discriminating on *treatment/control* are shown in Table 3, while sample sizes on *sex* are shown in Figure 1.

Figure 2 displays the age frequency distributions for the total sample (a) and both groups (b),(c). Skewness in (b) and (c) suggest higher concentration of younger people in control group compared to treatment.

Figure 3 shows AS4 (number of doses of morphine sulphate after operation), presents *right-skewed* histograms for all groups. Figures 3(b) and 3(c) suggest that treatment reduces average morphine usage.

Treatment group has (Figure 4) a distribution concentrated on lower pain scores. Control group has wider and higher range of scores. Treatment significantly lowers pain. Plotting histograms for VD4 in Figure 5.

1.2 Considerations on assignment mechanism

There is evidence (Table 4) of *moderate imbalance* in ages of *control* and *treatment* units. Table 5 shows proportions of males and females appear very similar among treatment and control units. Despite age shows some degree of imbalance, the sex variable appears to be well balanced across groups. On these results we **cannot** conclude that the mechanism **was not** random.

2 Fisher

Fisher’s exact p-value, would require computing *test statistic* on each realization of the *assignment vector*. This would lead to $\binom{N}{N_t} = 1.037199 \times 10^{17}$ different scenarios. This is unfeasible so we used *MonteCarlo approximation* using **difference in mean statistic** $T^{dif} = \bar{Y}_t^{obs} - \bar{Y}_c^{obs} = -21.607$ using different values of K (sample cardinality):

Table 8 shows $\hat{p}_{FRT}$ for each K , *null hypothesis* of *No effect whatsoever* can be rejected. The **asymptotic p-value** is $8.603783e - 05$, confirming the *rejection of null hypothesis*.

Results show a robust, significantly negative effect (90% CI: $[-10.0, -7.3]$), confirmed across Monte Carlo tests.

Table 9 show the setup for calculate the Confidence Interval. We employed a **model output** strategy based on a linear regression of the outcome on treatment assignment, and covariates and took the estimated treatment coefficient as test statistic which is $data\$W = -21.2901$ and statistically significant. Again using different samples cardinality, table 6 shows that, the estimated p-values confirms to be near zero (around 7.6×10^{-5}), confidently rejecting the null hypothesis.

Follow the previous method we calculate the 90% Fisher confidence interval, for constant additive treatment effect. We test a range of additive constant treatment effect τ , so that result begins to differ from noise, therefore, that value is considered plausible. The final range, that is, the set of test values that have not been refused, is $[-23.29, -19.29]$, with an observed effect of -21.29 , which confirms a reductive effect.

3 Neyman

3a. Estimate the ATE The objective is to estimate the Average Treatment Effect (ATE) for the target population. Under a CRE, the difference in observed sample means $\hat{\tau}$ is an unbiased estimator of the true population ATE. We see

that $\hat{\tau} = -21.607$, with an estimated standard error of 4.789. Therefore, our estimate suggests that the treatment has, on average in the sample, a reducing effect on the outcome of interest (VD4). The standard error has been estimated as $\sqrt{\hat{V}_{Neyman}}$.

3b and 3c. CI and Hypothesis Testing In 3b, we construct a 90% large-sample CI for the ATE, constructed using the asymptotic distribution of $\frac{\hat{\tau} - \tau}{\sqrt{V_W[\hat{\tau}]}}$ under randomization (which is $\mathcal{N}(0,1)$), and then approximating

the true standard error $\sqrt{V_W[\hat{\tau}]}$ with $\sqrt{\hat{V}_{Neyman}}$. The CI is $[-29.460, -13.754]$ (actual coverage likely $> 90\%$), showing the ATE is significantly non-zero. Testing $H_0 : \text{ATE} = 0$ is done with the statistic $\frac{\hat{\tau}}{\sqrt{\hat{V}_{Neyman}}}$ which, under H_0 and

for a large N , has approximately a standard normal distribution. This yields a two-sided p-value of $p = 6.415 \times 10^{-6}$, suggesting rejection of H_0 at conventional significance levels.

3d. Lin’s Estimator In point 3d, we tried to incorporate covariates into our inference. By computing Lin’s estimator, we get an ATE estimate of -21.315 . By calculating the *HC2* robust estimate of its variance, we get a value of 27.987, resulting in an estimated standard error of 5.290.

This shows that by adjusting our analysis to include covariates, we actually incur a loss of precision in our inference compared to the simple $\hat{\tau}$ estimator. A reason for this could be the small sample size used.

Table 10 report a comparison between results obtained with Neyman and Lin’s estimators.

4 Fisher SRE

This exercise performs a stratified Fisher test to isolate the treatment effect from gender-based differences. The method calculates a weighted difference-in-means statistic for women and men, respecting the dimensionality of each group. The null distribution is generated by constrained permutations, which reshuffle the treatment only within each stratum to maintain the group composition. The resulting p-values (Table 11) confirm the rejection of the null hypothesis.

5 Neyman and Lin SRE

To calculate the ATE assuming data from an SRE based on sex, we take two approaches. In the Neyman approach, we compute $\hat{\tau}_j$ within each stratum j and then pool them together based on the proportion of the population each stratum represents. The Neyman estimate of variance is similarly computed for

each stratum and then pooled together, resulting in a conservative estimate of the true variance of our estimator.

For Lin’s estimator, we again compute $\hat{\tau}_{int,j}^{ols}$ and the *HC0* (EHW variance estimator) estimate within each stratum, and then combine the stratum specific results based on the proportion of the population each stratum represents. Results are shown in Table 12. Both methods confirm that the treatment ($W = 1$) has a strong reducing effect on the outcome variable (VD4).

6 Table and figures

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Table 1: Statistics on numerical data

Variable name	Type	Mean	Standard Deviation	Variance	Max	Min
age	numerical	66.45	9.241	85.404	79	39
as4	numerical	13.433	9.866	97.334	44	1
vs4	numerical	36.083	17.733	314.485	80	5
vd4	numerical	55.083	21.263	452.111	100	10

Table 2: Statistics on categorical data

Total units	Total females	Total males
60	29	31

Table 3: Comparative Descriptive Statistics: treatment vs. Control Groups

Variable	treatment Group		Control Group	
	Mean	Standard deviation	Mean	Standard deviation
Age	68.188	8.364	64.464	9.931
AS4	11.500	9.748	15.643	9.700
VS4	27.969	12.754	45.357	18.254
Sample Size	N = 32 (15F, 17M)		N = 28 (14F, 14M)	

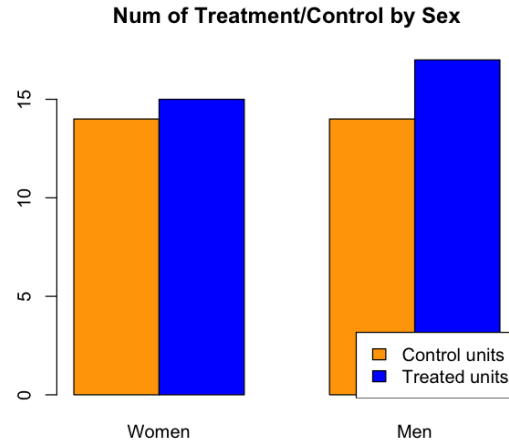


Figure 1: Distribution of treatment and Control groups by Sex.

Table 4: Mean age comparison

$\bar{X}_{Total}$	$\bar{X}_{age(0)}$	$\bar{X}_{age(1)}$	$\bar{X}_{age(1)} - \bar{X}_{age(0)}$
66.450	64.464	68.188	3.724

Table 5: Gender proportions comparison and Balance check

Proportion of gender in:	Total	Control	treatment
Men	0.517	0.500	0.531
Women	0.483	0.500	0.469

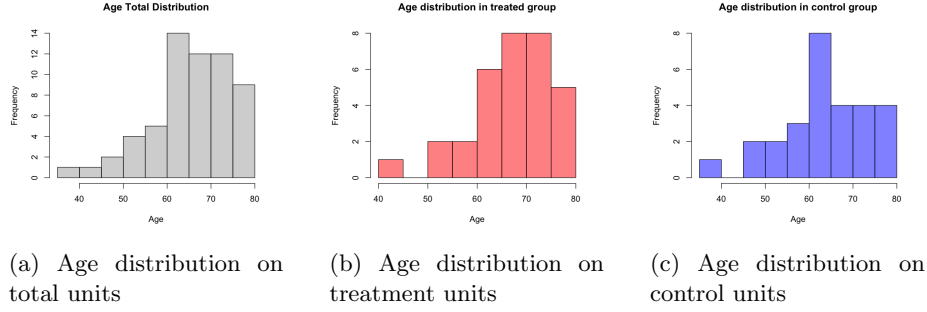


Figure 2: Age frequency distributions.

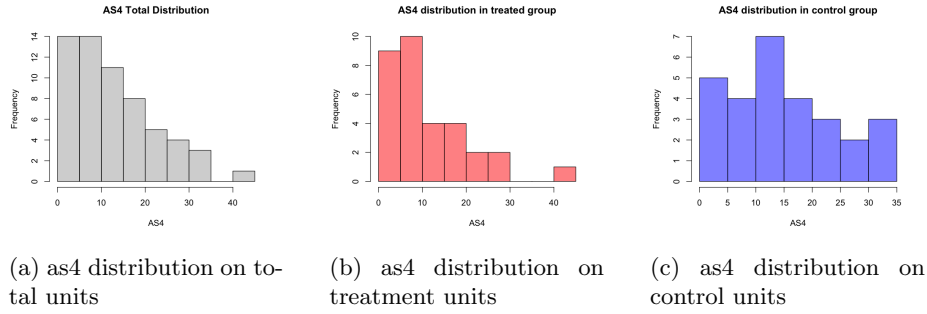


Figure 3: AS4 frequency distributions.

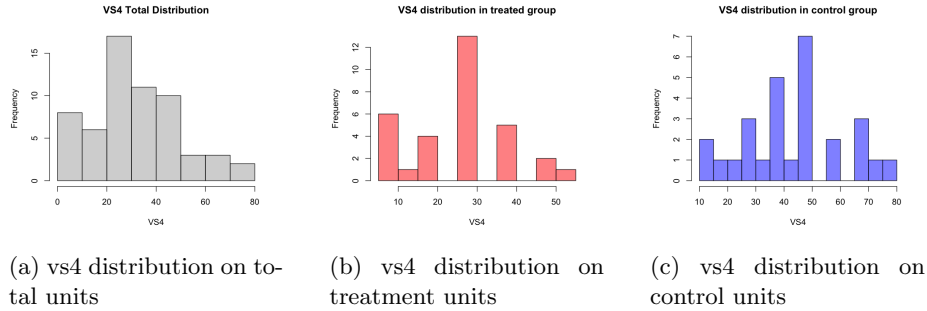
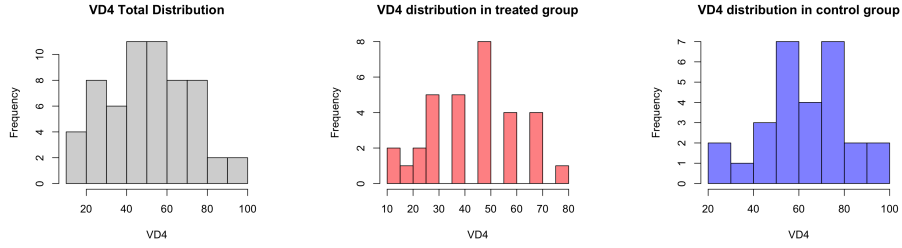


Figure 4: VS4 frequency distributions.

Table 6: Model Output strategy Monte Carlo approximations of the Fisher randomization test p-value for different numbers of replications K

K	$data\$W$
10000	1.0×10^{-4}
100000	6.0×10^{-5}
1000000	7.6×10^{-5}



(a) vd4 distribution on total units (b) vd4 distribution on treatment units (c) vd4 distribution on control units

Figure 5: VD4 frequency distributions.

Table 7: Mean age by gender and treatment status

	Control (0)	treatment (1)	$\bar{X}_{age,sex}(1) - \bar{X}_{age,sex}(0)$
Men	66.000	68.471	2.471
Women	62.929	67.867	4.938

Table 8: Estimated FRT p-values for different values of K

K	$\hat{p}_{FRT}$
10000	1.0×10^{-4}
100000	3.0×10^{-5}
1000000	3.7×10^{-5}

K	Confidence Level	Upper Bound	Lower Bound	τ
10000	90%	-7.3	-10	$[-10, 10]$

Table 9: Setup CI on Fisher

Estimator	Estimate	$s\hat{e}$
$\hat{\tau}$	-21.607	4.789
$\hat{\tau}_{int}^{ols}$	-21.315	5.290

Table 10: Comparison between Neyman and Lin's Estimators assuming data from a CRE (Exercise 3)

MonteCarlo iteration	p_value
1×10^4	2×10^{-4}
1×10^5	7×10^{-5}
1×10^6	2.5×10^{-5}

Table 11: Result of the Exercise 5

Estimator	Estimate	$\hat{s.e}$
$\hat{\tau}$	-21.611	4.848
$\hat{\tau}_{int}^{ols}$	-21.092	5.043

Table 12: Comparison between Neyman and Lin's Estimators assuming data from SRE (Exercices 6,7)